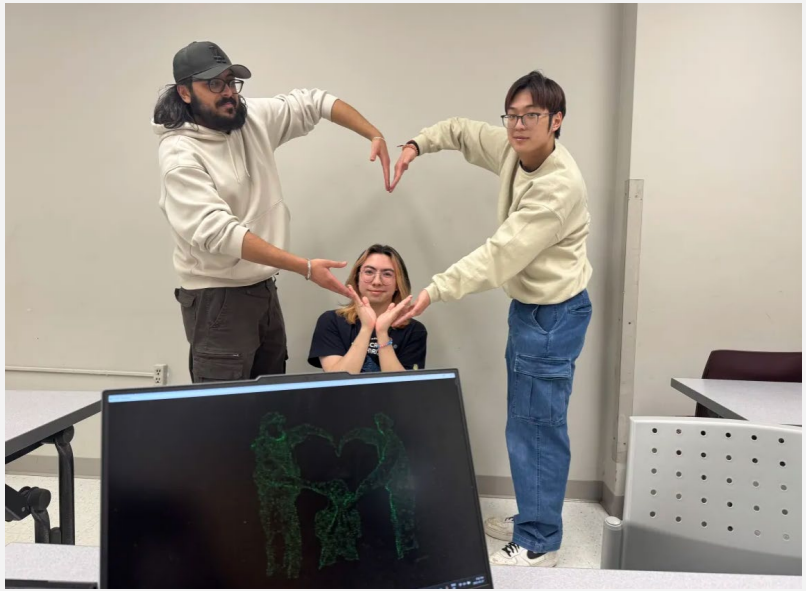
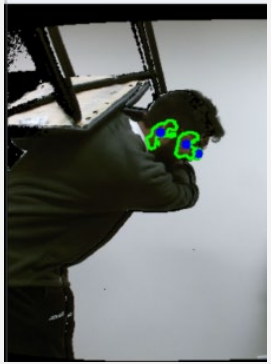
P.C.D.

Partially Covered Detection
(Team 14)





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01

Problem Statement

Problem Statement

Our project aims to design software that can reliably detect people using point cloud data. Specifically, our product will be able to accurately and consistently detect people whether they are visually obstructed or not.

Use Cases



Robotics Research

Furthers research in robot and human interactions



Assistive Robots

Assist humanitarian aid and domestic robot development

Requirements/Constraints

Functional

- Human Detection
- Location Estimation *
- Offline Processing
- Body Pose Variation Handling
- Integration with Kinect Sensor
- Real-Time Processing

Nonfunctional

- Usability *
- Reliability
- Accuracy

Constraints

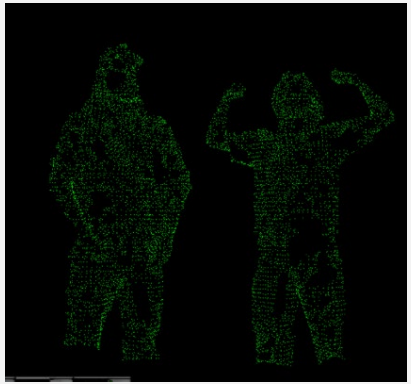
- No Machine Learning
- Using Kinect Sensor



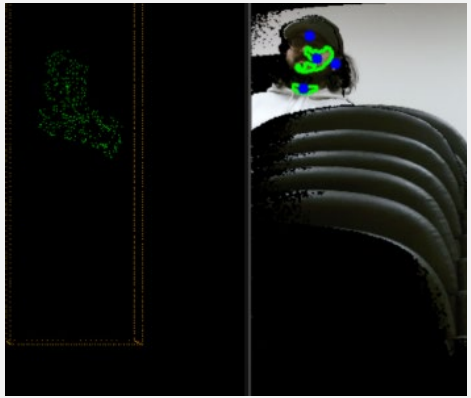
02

Live Demo

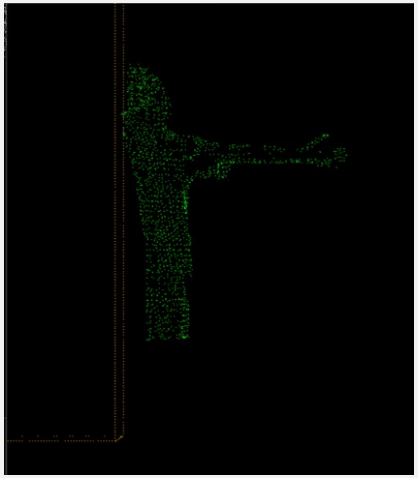
Demo



Fully Visible



Partially Obstructed



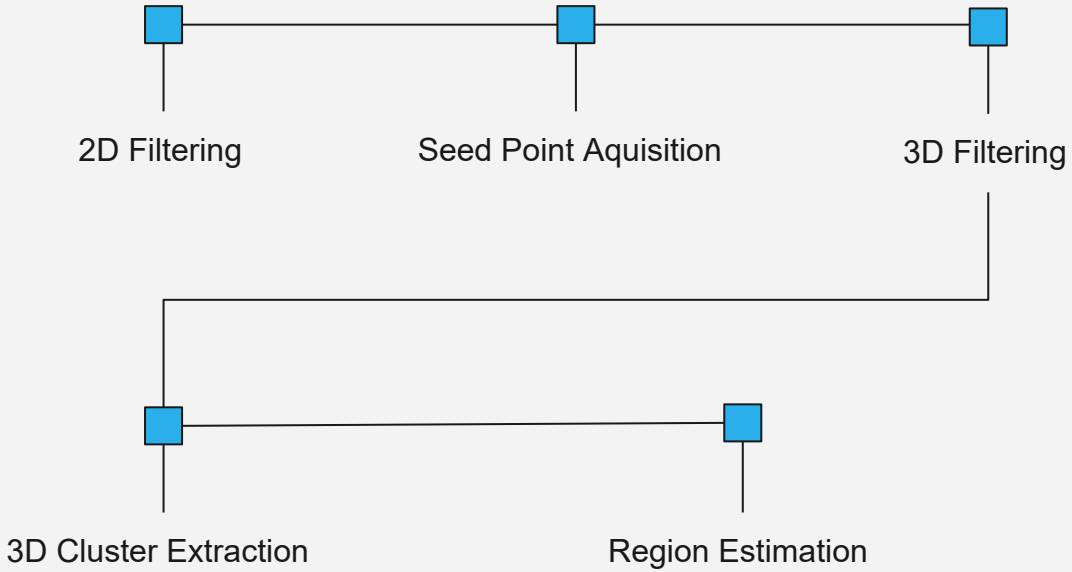
Partially Out of Frame



03

Methodology

Process Overview



2D Filtering

- Done in OpenCV
- Custom Plane Detection
- Plane Filtering/Removal





Seed Point Acquisition

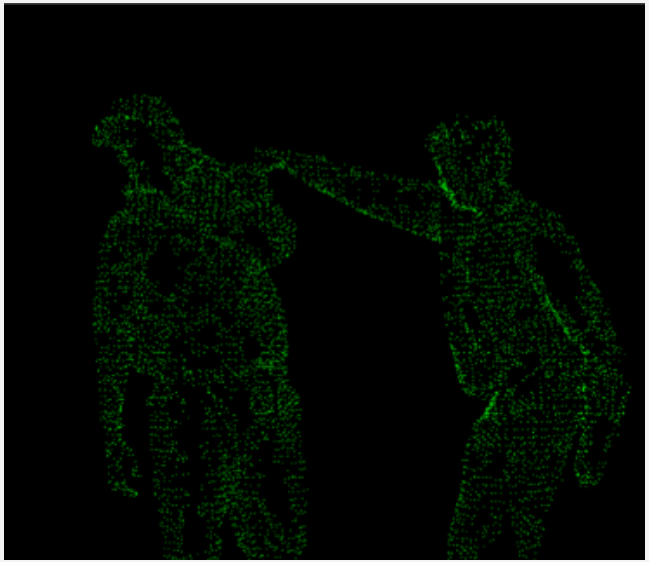
- Done in OpenCV
- Skin Point Detection
    - Initial Skin Mask
    - Skin Mask Filtering
    - Skin Confidence Mask

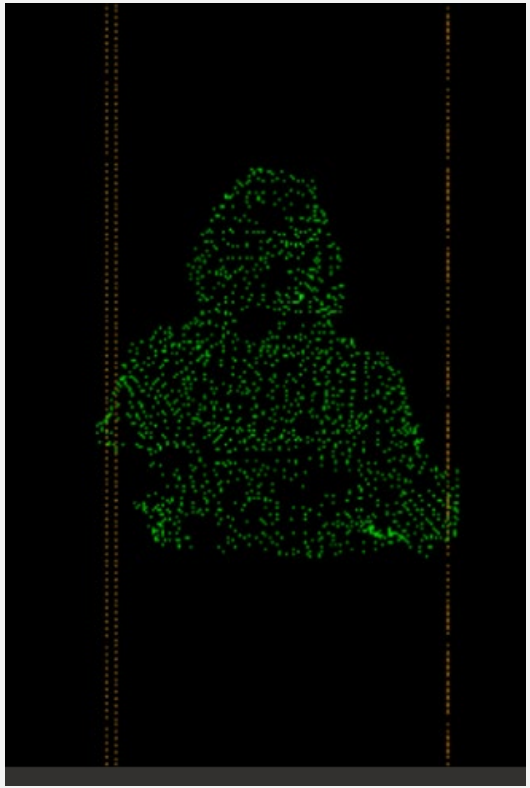
3D Filtering

- Done in PCL
- Voxel Down Sampling
- Noise Removal
- PCL Plane Removal (RANSAC)

3D Cluster Extraction

- Start from Detected Seed Points
- Region Growing From Seed Point
    - Conditional On:
        - Max Neighbour Distance
        - Growth Radius
- Track Growth Directions
- Enhance Cluster Resolution





Region Estimation

- Cluster Neighbor Centroids
- Compute Growth Direction/Clustering Pattern
- Determine Region Centroid/Offset



04

Next Steps

Body Part Segmentation

- Specific Body Part Identification
- Refine Region Estimation



Robot Integration

- Adapting for Different Environments
- Adapt Sensor to Robot Input
- Output to Robot Action



Improvements

- More Detection Methods
- Reduce False Positives
- Reliability/Consistency





05

Questions?

Thank You!